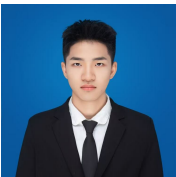


Tenghui Tu

English names: Titanfy / Tevin Tu
Embedded Development · Computer Vision · HMI & System Integration
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EDUCATION

Hunan Agricultural University Sep 2024–Present
Xiangjiang College of Engineering Excellence · Undergraduate student

SKILLS

Embedded systems & control: C, Python, RTOS; STM32F4, Raspberry Pi 5B, UART/USART, bus servos; active disturbance rejection control (ADRC) and sensor fusion.

Vision & edge computing: OpenCV; YOLOv8n edge deployment; interface testing and debugging for MediaPipe Pose estimation and YOLO-Face + ArcFace face recognition.

HMI & system integration: TJC serial touchscreens, USART HMI; UI interactions, serial protocols, state synchronization, hardware integration and exception testing.

Mechanical design & simulation: SolidWorks, structural modeling, transmission mechanisms and PCB design; basic familiarity with ROS, Gazebo simulation and algorithm validation workflows.

Industrial vision & PLC: Hikvision industrial cameras, image acquisition and processing; basic knowledge of PLCs and industrial vision/control integration.

Digital media, photography & AIGC: Photography, short-film production, AI-generated video; Photoshop, After Effects, Premiere Pro; content planning and operations for Xiaohongshu and Douyin.

Productivity & AI tools: Word, Excel; Codex, Claude AI, Doubao, Hermes and OpenClaw for development assistance and content creation.

Leadership & collaboration: Lead the tabletop gaming project across product planning and hardware/software implementation; coordinate tasks, technical communication and issue validation in competition practice and internship integration work.

INTERNSHIP EXPERIENCE

Changsha Kangtong Electronic Technology Co., Ltd. Aug 2026–Present
AI Fitness Assessment Device | Algorithm Interface Debugging & Hardware Integration

- Test and debug interfaces for an RK3588-based face-recognition pipeline combining YOLO-Face detection with ArcFace feature extraction and matching; integrate camera hardware and validate recognition outputs.
- Test MediaPipe Pose interface outputs and integrate devices for push-ups, standing long jumps, sit-ups, jump rope and pull-ups in automated fitness-assessment scenarios.
- Work with product, development and testing teams to clarify recognition issues and functional requirements; reproduce, diagnose and validate issues during algorithm-interface and device integration.

Shenzhen Weier Technology Jul–Aug 2026
Cotton-Topping Robot | Embedded Control & Field Integration

- Contributed to STM32F4 cutting-sequence analysis and optimization, examining timing relationships between visual target detection, wheel-odometry/IMU data and blade-lowering triggers.
- Traveled to Xinjiang for field hardware integration and testing; analyzed vehicle motion, actuator response, missed cuts and unintended plant damage, and worked with engineers to optimize and validate cutting timing.

Shenzhen InnoX Academy Jan 2026
Robotics R&D Department | HMI / Host-Application Software Engineering Intern

- Independently developed a TJC serial-touchscreen HMI from requirements and page architecture through UI construction and interaction logic, delivering operation controls, parameter settings, status displays and alarms.
- Implemented page navigation, button responses and variable updates in USART HMI; independently designed the serial communication protocol following team discussions, defining control commands, status feedback and data fields for parameter transmission and state synchronization.
- Worked with the controller team to resolve command-parsing, data-refresh and page-synchronization issues; completed communication exception tests, compilation, flashing and on-device validation, delivering a deployable HMI project.

PROJECTS

Intelligent Tobacco-Stalk Clearing Machine | Mechanical & Electrical Control Lead

- Designed overall structures, transmission mechanisms and cutter rollers; iterated low-level drive control and clearing-path designs.
The project secured RMB 50,000 in investment from the Golden Seed Program and received coverage from People's Daily and Hunan Satellite TV.

Face-Vision Access Control | Edge Vision & Actuator Integration

- Deployed YOLOv8n on a Raspberry Pi 5B for face detection and real-time vision processing; drove a servo over bus communication to link visual outputs with access-barrier movement.

Interactive Tabletop Gaming Device | Project Lead

- Selected hardware, designed the PCB, and implemented speaker-drive and LED-strip control logic; contributed to a host-side software agent linking game phases with audio and lighting feedback.
- Led product planning and hardware/software collaboration; the project received recognition from Xiangjiang Science and Technology Innovation Institute and an opportunity to be showcased at the institute.

Electronics Design Competition Practice | Control Architecture & Motion Control

- Designed the electrical control architecture and implemented ADRC, sensor fusion, line-following and obstacle-avoidance logic, collaborating with the team on embedded development.

CAMPUS EXPERIENCE

Nongmou Alvideo Team | Team Member

- Produce AI-generated promotional videos for agricultural machinery, translating agricultural technology into audiovisual content and generating profit through the related video business.

Agricultural Artificial Intelligence Laboratory, Hunan Agricultural University Dec 2024–Present
Undergraduate Member | Tabletop Gaming Product Project Lead

- Contribute to agricultural AI research and development; lead the tabletop gaming product track, coordinating product design, hardware/software implementation and project collaboration.

Xiangjiang College of Engineering Excellence | Academic Affairs Assistant Sep 2025–Present

- Support teaching administration and student affairs; help build and maintain the online teaching platform, enter course information, organize learning resources and communicate notices.

Smart Agriculture Association, Hunan Agricultural University | Vice President Dec 2024–Present

- Define departmental responsibilities and coordinate cross-department collaboration and daily work to support association operations and team execution.

Summer Rural Outreach Program | Yongzhou Field Team Member Aug 2025

- Conducted field research, outreach, data collection and documentation; worked with the team on activity planning, site visits, delivery and final reporting.